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References Book:

1. **Jiawei Han and Micheline Kamber, Data Mining: Concepts and Techniques, Morgan Kaufmann Publishers, third edition ,2013**
2. **Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar, Introduction to Data Mining, second edition, Pearson, 2019**
3. **Ian. H. Witten, Eibe Frank and Mark. A. Hall, Data Mining: Practical Machine Learning Tools and Techniques, third edition , 2017**

12/12/2023

FACULTY **HOD** **DEAN ACADEMICS**

UNIT IV

Cluster Analysis :

What is Cluster Analysis?

- **Cluster: a collection of data objects**
  - **Similar to one another within the same cluster**
  - **Dissimilar to the objects in other clusters**
- **Cluster analysis**
  - **Grouping a set of data objects into clusters**
- **Clustering is unsupervised classification: no predefined classes**
- **Typical applications**
  - **As a stand-alone tool to get insight into data distribution**
  - **As a preprocessing step for other algorithms**

General Applications of Clustering :

- **Pattern Recognition**
- **Spatial Data Analysis**
  - **create thematic maps in GIS by clustering feature spaces**
  - **detect spatial clusters and explain them in spatial data mining**
- **Image Processing**
- **Economic Science (especially market research)**
- **WWW**
  - **Document classification**
  - **Cluster Weblog data to discover groups of similar access patterns**

Examples of Clustering Applications :

- **Marketing: Help marketers discover distinct groups in their customer bases,and then use this knowledge to develop targeted marketing programs**
- **Land use: Identification of areas of similar land use in an earth observationdatabase**
- **Insurance: Identifying groups of motor insurance policy holders with a highaverage claim cost**

- **City-planning: Identifying groups of houses according to their house type,value, and geographical location**
- **Earth-quake studies: Observed earth quake epicenters should be clusteredalong continent faults**

What Is Good Clustering?

- **A good clustering method will produce high quality clusters with**
  - **high intra-class similarity**
  - **low inter-class similarity**
- **The quality of a clustering result depends on both the similarity measure usedby the method and its implementation.**
- **The quality of a clustering method is also measured by its ability to discoversome or all of the hidden patterns.**

Requirements of Clustering in Data Mining :

- **Scalability**
- **Ability to deal with different types of attributes**
- **Discovery of clusters with arbitrary shape**
- **Minimal requirements for domain knowledge to determine input parameters**
- **Able to deal with noise and outliers**
- **Insensitive to order of input records**
- **High dimensionality**
- **Incorporation of user-specified constraints**
- **Interpretability and usability**

Type of data in clustering analysis :

- **Interval-scaled variables:**
- **Binary variables:**
- **Nominal, ordinal, and ratio variables:**
- **Variables of mixed types:**

Types Of Data Structures :

First of all, let us know what types of data structures are widely used in cluster analysis.

We shall know the types of data that often occur in cluster analysis and how to preprocess them for such analysis.

Suppose that a data set to be clustered contains n objects, which may represent persons, houses, documents, countries, and so on.

Main memory-based clustering algorithms typically operate on either of the following two data structures.

Types of data structures in cluster analysis are :

- **Data Matrix (or object by variable structure)**
- **Dissimilarity Matrix (or object by object structure)**

Data Matrix :

This represents n objects, such as persons, with p variables (also called measurements or attributes), such as age, height, weight, gender, race and so on. The structure is in the form of a relational table, or n-by-p matrix (n objects x p variables)

The Data Matrix is often called a two-mode matrix since the rows and columns of this represent the different entities.

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

Dissimilarity Matrix:

This stores a collection of proximities that are available for all pairs of n objects. It is often represented by a n – by – n table, where d(i,j) is the measured difference or dissimilarity between objects i and j. In general, d(i,j) is a non-negative number that is close to 0 when objects i and j are higher similar or "near" each other and becomes larger the more they differ. Since d(i,j) = d(j,i) and d(i,i) =0, we have the matrix in figure.

This is also called as one mode matrix since the rows and columns of this represent the same entity.

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \ldots & \ldots & 0 \end{bmatrix}$$

Types Of Data In Cluster Analysis Are :

Interval-Scaled Variables :

Interval-scaled variables are continuous measurements of a roughly linear scale.

Typical examples include weight and height, latitude and longitude coordinates (e.g., when clustering houses), and weather temperature.

The measurement unit used can affect the clustering analysis. For example, changing measurement units from meters to inches for height, or from kilograms to pounds for weight, may lead to a very different clustering structure.

In general, expressing a variable in smaller units will lead to a larger range for that variable, and thus a larger effect on the resulting clustering structure.

To help avoid dependence on the choice of measurement units, the data should be standardized. Standardizing measurements attempts to give all variables an equal weight.

This is especially useful when given no prior knowledge of the data. However, in some applications, users may intentionally want to give more weight to a certain set of variables than to others.

For example, when clustering basketball player candidates, we may prefer to give more weight to the variable height.

Binary Variables :

A binary variable is a variable that can take only 2 values.

For example, generally, gender variables can take 2 variables male and female.

Contingency Table For Binary Data:

Let us consider binary values 0 and 1

	1	0	*sum*
1	a	b	$a+b$
0	c	d	$c+d$
sum	$a+c$	$b+d$	p

Let p=a+b+c+d

Simple matching coefficient (invariant, if the binary variable is symmetric):

$$d(i, j) = \frac{b + c}{a + b + c + d}$$

Jaccard coefficient (noninvariant if the binary variable is asymmetric):

$$d(i, j) = \frac{b + c}{a + b + c}$$

Nominal or Categorical Variables :

A generalization of the binary variable in that it can take more than 2 states, e.g., red, yellow, blue, green.

Method 1: Simple matching:

The dissimilarity between two objects i and j can be computed based on the simple matching.

m: Let m be no of matches (i.e., the number of variables for which i and j are in the same state).

p: Let p be total no of variables.

$$d(i, j) = \frac{p - m}{p}$$

Method 2: use a large number of binary variables:

Creating a new binary variable for each of the M nominal states.

Ordinal Variables :

An ordinal variable can be discrete or continuous.

In this order is important, e.g., rank.

It can be treated like interval-scaled

By replacing xif by their rank,

$$r_{if} \in \{1, \ldots, M_f\}$$

By mapping the range of each variable onto [0, 1] by replacing the i-th object in the f-th variable by,

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

Then compute the dissimilarity using methods for interval-scaled variables.

Ratio-Scaled Intervals :

Ratio-scaled variable: It is a positive measurement on a nonlinear scale, approximately at an exponential scale, such as Ae^Bt or A^e-Bt.

Methods:

- **First, treat them like interval-scaled variables — not a good choice! (why?)**
- **Then apply logarithmic transformation i.e. y = log(x)**

- **Finally, treat them as continuous ordinal data treat their rank as interval-scaled.**

Variables Of Mixed Type :

A database may contain all the six types of variables
symmetric binary, asymmetric binary, nominal, ordinal, interval, and ratio.

And those combinedly called as mixed-type variables.

A Categorization of Major Clustering Methods:

1. **Partitioning algorithms: Construct various partitions and then evaluate themby some criterion**
2. **Hierarchy algorithms: Create a hierarchical decomposition of the set of data(or objects) using some criterion**
3. **Density-based: based on connectivity and density functions**
4. **Grid-based: based on a multiple-level granularity structure**
5. **Model-based: A model is hypothesized for each of the clusters and the idea isto find the best fit of that model to each other**

- **Partitioning Algorithms: Basic Concept:**

  Partitioning method:

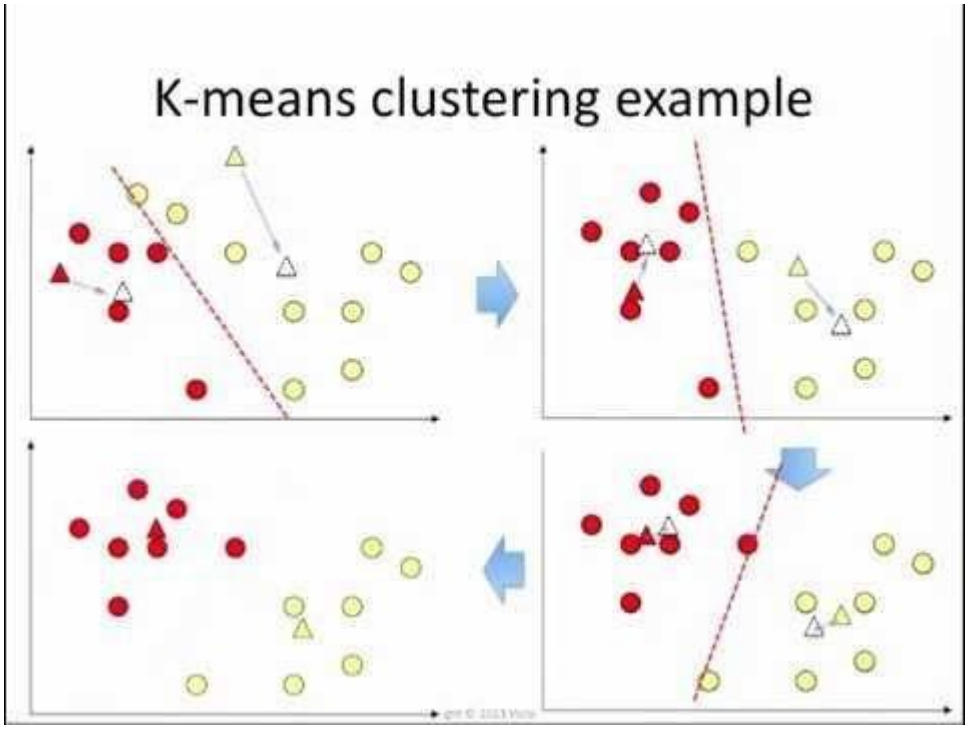
- **Construct a partition of a database D of n objects into a set of k clusters**
- **Given a k, find a partition of k clusters that optimizes the chosen partitioning criterion:**
  - **Global optimal: exhaustively enumerate all partitions**
  - **Heuristic methods: k-means and k-medoids algorithms**
  - **k-means (MacQueen'67): Each cluster is represented by the center of the cluster**
  - **k-medoids or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster**

The K-Means Clustering Method

- **Given k, the k-means algorithm is implemented in 4 steps:**
  - **Partition objects into k nonempty subsets**
  - **Compute seed points as the centroids of the clusters of the currentpartition. The centroid is the center (mean point) of the cluster.**
  - **Assign each object to the cluster with the nearest seed point.**
  - **Go back to Step 2, stop when no more new assignment.**

The K-Means Clustering Method:

- **Example :**



Comments on the K-Means Method :

- **Strength**
  - **Relatively efficient: O(tkn), where n is # objects, k is # clusters, and t is # iterations. Normally, k, t << n.**
  - **Often terminates at a local optimum. The global optimum may be found using techniques such as: deterministic annealing and genetic algorithms**

- **Weakness**
  - **Applicable only when mean is defined, then what about categoricaldata?**
  - **Need to specify k, the number of clusters, in advance**
  - **Unable to handle noisy data and outliers**
  - **Not suitable to discover clusters with non-convex shapes**

Variations of the K-Means Method:

- **A few variants of the k-means which differ in**
  - **Selection of the initial k means**
  - **Dissimilarity calculations**
  - **Strategies to calculate cluster means**
- **Handling categorical data: k-modes (Huang'98)**
  - **Replacing means of clusters with modes**
  - **Using new dissimilarity measures to deal with categorical objects**
  - **Using a frequency-based method to update modes of clusters**
  - **A mixture of categorical and numerical data: k-prototype method**

The K-Medoids Clustering Method:

- **Find representative objects, called medoids, in clusters**
- **PAM (Partitioning Around Medoids, 1987)**
  - **starts from an initial set of medoids and iteratively replaces one of the medoids by one of the non-medoids if it improves the total distance of the resulting clustering**
  - **PAM works effectively for small data sets, but does not scale well for large data sets**
- **CLARA (Kaufmann & Rousseeuw, 1990)**
- **CLARANS (Ng & Han, 1994): Randomized sampling**
- **Focusing + spatial data structure (Ester et al., 1995)**

K-Medoids Clustering Method :

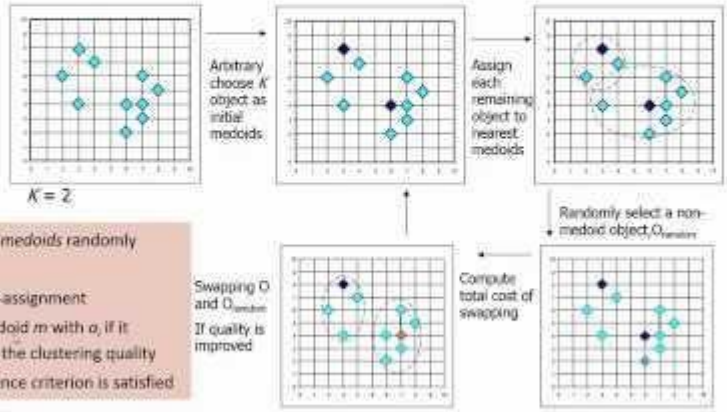
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PAM (Partitioning Around Medoids) (1987) :

- **PAM (Kaufman and Rousseeuw, 1987), built in Splus**
- **Use real object to represent the cluster**
  - **Select k representative objects arbitrarily**
  - **For each pair of non-selected object h and selected object i, calculatethe total swapping cost TCih**
  - **For each pair of i and h,**
    - **If TCih < 0, i is replaced by h**
    - **Then assign each non-selected object to the most similar representative object**
  - **repeat steps 2-3 until there is no change**

PAM Clustering: Total swapping cost

PAM: A Typical *K-Medoids* Algorithm



Select initial *K medoids* randomly

Repeat

Object re-assignment

Swap medoid *m* with o_i if it improves the clustering quality

Until convergence criterion is satisfied

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CLARA (Clustering Large Applications) (1990) :

- **CLARA (Kaufmann and Rousseeuw in 1990)**
  - **Built in statistical analysis packages, such as S+**
  - **It draws multiple samples of the data set, applies PAM on each sample, andgives the best clustering as the output**
- **Strength: deals with larger data sets than PAM**
- **Weakness:**
  - **Efficiency depends on the sample size**
  - **A good clustering based on samples will not necessarily represent agood clustering of the whole data set if the sample is biased**

CLARANS ("Randomized" CLARA) (1994) :

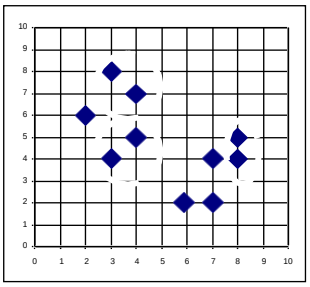
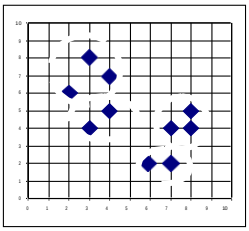
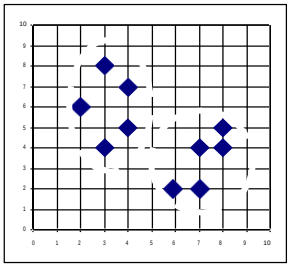
- **CLARANS (A Clustering Algorithm based on Randomized Search) (Ng andHan'94)**
- **CLARANS draws sample of neighbors dynamically**
- **The clustering process can be presented as searching a graph where everynode is a potential solution, that is, a set of k medoids**
- **If the local optimum is found, CLARANS starts with new randomly selectednode in search for a new local optimum**
- **It is more efficient and scalable than both PAM and CLARA**
- **Focusing techniques and spatial access structures may further improve itsperformance (Ester et al.'95)**

Hierarchical Clustering :

Use distance matrix as clustering criteria. This method does not require the numberof clusters k as an input, but needs a termination condition

AGNES (Agglomerative Nesting)

- **Introduced in Kaufmann and Rousseeuw (1990)**
- **Implemented in statistical analysis packages, e.g., Splus**
- **Use the Single-Link method and the dissimilarity matrix.**
- **Merge nodes that have the least dissimilarity**
- **Go on in a non-descending fashion**
- **Eventually all nodes belong to the same cluster**

A Dendrogram Shows How the Clusters are Merged Hierarchically:

Decompose data objects into a several levels of nested partitioning (tree of clusters),called a dendrogram.

A clustering of the data objects is obtained by cutting the dendrogram at the desiredlevel, then each connected component forms a cluster.

DIANA (Divisive Analysis) :

- **Introduced in Kaufmann and Rousseeuw (1990)**
- **Implemented in statistical analysis packages, e.g., Splus**
- **Inverse order of AGNES**

- **Eventually each node forms a cluster on its own.**

More on Hierarchical Clustering Methods :

- **Major weakness of agglomerative clustering methods**
  - **do not scale well: time complexity of at least O(n2), where n is thenumber of total objects**
  - **can never undo what was done previously**
- **Integration of hierarchical with distance-based clustering**
  - **BIRCH (1996): uses CF-tree and incrementally adjusts the quality ofsub-clusters**
  - **CURE (1998): selects well-scattered points from the cluster and thenshrinks them towards the center of the cluster by a specified fraction**
  - **CHAMELEON (1999): hierarchical clustering using dynamicmodeling**

7.5.2 BIRCH (1996) :

- **Birch: Balanced Iterative Reducing and Clustering using Hierarchies, by Zhang, Ramakrishnan, Livny (SIGMOD'96)**
- **Incrementally construct a CF (Clustering Feature) tree, a hierarchical data structure for multiphase clustering**
  - **Phase 1: scan DB to build an initial in-memory CF tree (a multi-level compression of the data that tries to preserve the inherent clustering structure of the data)**
  - **Phase 2: use an arbitrary clustering algorithm to cluster the leaf nodes of the CF-tree**
- **Scales linearly: finds a good clustering with a single scan and improves the quality with a few additional scans**
- **Weakness: handles only numeric data, and sensitive to the order of the data record.**

Rock Algorithm and CHAMELEON:

- **ROCK: Robust Clustering using linKs,**
  by S. Guha, R. Rastogi, K. Shim (ICDE'99).
  - **Use links to measure similarity/proximity**
  - **Not distance based**

- **Computational complexity:**

- **Basic ideas:**
  - **Similarity function and neighbors:Let T1 = {1,2,3}, T2={3,4,5}**

- **Rock: Algorithm:**

Links: The number of common neighbours for the two points.
{1,2,3}, {1,2,4}, {1,2,5}, {1,3,4}, {1,3,5}

{1,4,5}, {2,3,4}, {2,3,5}, {2,4,5}, {3,4,5}

{1,2,3} 3 {1,2,4}



- **Algorithm**
  - **Draw random sample**
  - **Cluster with links**
  - **Label data in disk**
  -

CHAMELEON:

- **CHAMELEON: hierarchical clustering using dynamic modeling, by G. Karypis, E.H. Han and V. Kumar'99**
- **Measures the similarity based on a dynamic model**
  - **Two clusters are merged only if the interconnectivity and closeness (proximity) between two clusters are high relative to the internal interconnectivity of the clusters and closeness of items within theclusters**
- **A two phase algorithm**
  - **1. Use a graph partitioning algorithm: cluster objects into a large number of relatively small sub-clusters**
  - **2. Use an agglomerative hierarchical clustering algorithm: find the genuine clusters by repeatedly combining these sub-clusters**

AGGLOMERATIVE HIERARCHICAL CLUSTERING :

Algorithms of hierarchical cluster analysis are divided into the two

categories divisible algorithms and agglomerative algorithms. A divisible algorithm starts from the entire set of samples X and divides it into a partition of subsets, then divides each subset into smaller sets, and so on. Thus, a divisible algorithm generates a sequence of partitions that is ordered from a coarser one to a finer one. An agglomerative algorithm first regards each object as an initial cluster. The clusters are merged into a coarser partition, and the merging process proceeds until the trivial partition is obtained: all objects are in one large cluster. This process of clustering is a bottom-up process, where partitions from a finer one to a coarser one.

Most agglomerative hierarchical clustering algorithms are variants of the single-link or complete-link algorithms. In the single-link method, the distance between two clusters is the minimum of the distances between all pairs of samples drawn from the two clusters (one element from the first cluster, the other from the second). In the complete-link algorithm, the distance between two clusters is the maximum of all distances between all pairs drawn from the two clusters. A graphical illustration of these two distance measures is given.

The basic steps of the agglomerative clustering algorithm are the same. Thesesteps are

1. **Place each sample in its own cluster. Construct the list of inter-cluster distances for all distinct unordered pairs of samples, and sort this list inascending order.**
2. **Step through the sorted list of distances, forming for each distinct threshold value d_k a graph of the samples where pairs samples closer than d_k are connected into a new cluster by a graph edge. If all the samples are members of a connected graph, stop. Otherwise, repeat this step.**
3. **The output of the algorithm is a nested hierarchy of graphs, which can be cut at the desired dissimilarity level forming a partition (clusters) identified by simple connected components in the corresponding sub graph.**

Let us consider five points {x_1, x_2, x_3, x_4, x_5} with the following coordinates as atwo-dimensional sample for clustering:

For this example, we selected two-dimensional points because it is easier to graphically represent these points and to trace all the steps in the clustering algorithm.

The distances between these points using the Euclidian

measure ared(x_1 , x_2) =2, d(x_1, x_3) = 2.5, d(x_1, x_4) = 5.39, d(x_1,

x_5) = 5

d(x_1 , x_3) =1.5, d(x_2, x_4) = 5, d(x_2, x_5) = 5.29,

d(x_3 , x_4) =3.5, d(x_3, x_5) = 4.03, d(x_4, x_5) = 2

The distances between points as clusters in the first iteration are the same for both single-link and complete-link clustering. Further computation for these twoalgorithms is different. Using agglomerative single-link clustering, the following steps are performed to create a cluster and to represent the cluster structure as adendrogram.

Hierarchical and Non-Hierarchical Clustering :

There are two main types of clustering techniques, those that create a hierarchy of clusters and those that do not. The hierarchical clustering techniquescreate a hierarchy of clusters from small to big. The main reason for this is that, aswas already stated, clustering is an unsupervised learning technique, and as such,there is no absolutely correct answer. For this reason and depending on the particularapplication of the clustering, fewer or greater numbers of clusters may be desired.With a hierarchy of clusters defined it is possible to choose the number of clustersthat are desired. At the extreme it is possible to have as many clusters as there arerecords in the database. In this case the records within the cluster are optimallysimilar to each other (since there is only one) and certainly different from the otherclusters. But of course such a clustering technique misses the point in the sense thatthe idea of clustering is to find useful patters in the database that summarize it andmake it easier to understand. Any clustering algorithm that ends up with as many clusters as there are records has not helped the user understand the data any better. Thus one of the main points about clustering is that there be many fewer clusters thanthere are original records. Exactly how many clusters should be formed is a matter ofinterpretation. The advantage of hierarchical clustering methods is that they allowthe end user to choose from either many clusters or

only a few.

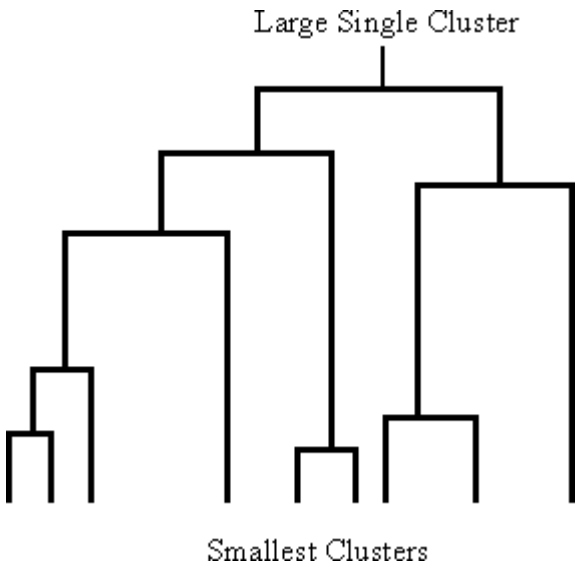
The hierarchy of clusters is usually viewed as a tree where the smallest clusters merge together to create the next highest level of clusters and those at that level merge together to create the next highest level of clusters. Figure 1.5 below shows how several clusters might form a hierarchy. When a hierarchy of clusters likethis is created the user can determine what the right number of clusters is that adequately summarizes the data while still providing useful information (at the other extreme a single cluster containing all the records is a great summarization but does not contain enough specific information to be useful).

This hierarchy of clusters is created through the algorithm that builds the clusters. There are two main types of hierarchical clustering algorithms:

- **Agglomerative - Agglomerative clustering techniques start with as many clusters as there are records where each cluster contains just one record. The clusters that are nearest each other are merged together to form the next largest cluster. This merging is continued until a hierarchy of clusters is built with just a single cluster containing all the records at the top of the hierarchy.**
- **Divisive - Divisive clustering techniques take the opposite approach from agglomerative techniques. These techniques start with all the records in one cluster and then try to split that cluster into smaller pieces and then in turn to try to split those smaller pieces.**

Of the two the agglomerative techniques are the most commonly used for clustering and have more algorithms developed for them. We'll talk about these in more detail in the next section. The non-hierarchical techniques in general are faster to create from the historical database but require that the user make some decision about the number of clusters desired or the minimum "nearness" required for two records to be within the same cluster. These non-hierarchical techniques often times are run multiple times starting off with some arbitrary or even random clustering and then iteratively improving the clustering by shuffling some records around. Or these techniques sometimes create clusters that are created with only one pass through the database adding records to existing clusters when they exist and creating new clusters when no existing

cluster is a good candidate for the given record. Because the definition of which clusters are formed can depend on these initial choices of which starting clusters should be chosen or even how many clusters these techniques can be less repeatable than the hierarchical techniques and can sometimes create either too many or too few clusters because the number of clusters is predetermined by the user not determined solely by the patterns inherent in the database.



Above figure Diagram showing a hierarchy of clusters. Clusters at the lowest levelare merged together to form larger clusters at the next level of the hierarchy.

Non-Hierarchical Clustering:

There are two main non-hierarchical clustering techniques. Both of them are very fast to compute on the database but have some drawbacks. The first are the single pass methods. They derive their name from the fact that the database must only be passed through once in order to create the clusters (i.e. each record is only read from the database once). The other class of techniques

are called reallocation methods. They get their name from the movement or "reallocation" of records from one cluster to another in order to create better clusters. The reallocation techniques do use multiple passes through the database but are relatively fast in comparison to the hierarchical techniques.

Hierarchical Clustering:

Hierarchical clustering has the advantage over non-hierarchical techniques in that the clusters are defined solely by the data (not by the users predetermining the number of clusters) and that the number of clusters can be increased or decreased by simple moving up and down the hierarchy.

The hierarchy is created by starting either at the top (one cluster that includes all records) and subdividing (divisive clustering) or by starting at the bottom with as many clusters as there are records and merging (agglomerative clustering). Usually the merging and subdividing are done two clusters at a time.

The main distinction between the techniques is their ability to favor long, scraggly clusters that are linked together record by record, or to favor the detection ofthe more classical, compact or spherical cluster that was shown at the beginning of this section. It may seem strange to want to form these long snaking chain like clusters, but in some cases they are the patters that the user would like to have detected in the database. These are the times when the underlying space looks quite different from the spherical clusters and the clusters that should be formed are not based on the distance from the center of the cluster but instead based on the records being "linked" together. Consider the example shown in Figure 1.6 or in Figure 1.7. In these cases there are two clusters that are not very spherical in shape but could be detected by the single link technique.

When looking at the layout of the data in Figure1.6 there appears to be two relatively flat clusters running parallel to each along the income axis. Neither the complete link nor Ward's method would, however, return these two clusters to the user. These techniques rely on creating a "center" for each cluster and picking these centers so that they average distance of each record from this center is minimized. Points that are very distant from these centers would necessarily fall into a different cluster.

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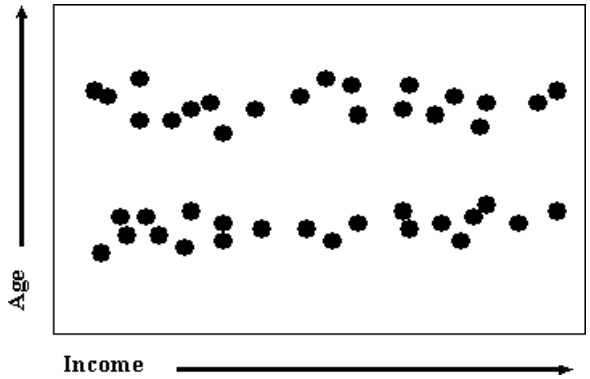


Figure 1.6 an example of elongated clusters which would not be recovered by the complete link or Ward's methods but would be by the single-link method.

Methods Density-Based Clustering:

- **Clustering based on density (local cluster criterion), such as density-connected points**
- **Major features:**
  - **Discover clusters of arbitrary shape**
  - **Handle noise**
  - **One scan**
  - **Need density parameters as termination condition**
- **Several interesting studies:**
  - **DBSCAN: Ester, et al. (KDD'96)**
  - **OPTICS: Ankerst, et al (SIGMOD'99).**
  - **DENCLUE: Hinneburg & D. Keim (KDD'98)**
  - **CLIQUE: Agrawal, et al. (SIGMOD'98)**

Density-Based Clustering: Background :

- **Two parameters:**
  - **Eps: Maximum radius of the neighbour hood**

- **MinPts: Minimum number of points in an Eps-neighbour hood of thatpoint**

- **NEps(p): {q belongs to D | dist(p,q) <= Eps}**
- **Directly density-reachable: A point p is directly density-reachable from apoint q wrt. Eps, MinPts if**
  - **1) p belongs to NEps(q)**
  - **2) core point condition:**

  |NEps (q)| >= MinPts

Density-Based Clustering: Background (II) :

- **Density-reachable:**
  - **A point p is density-reachable from a point q wrt. Eps, MinPts if there is a chain of points p1, …, pn, p1 = q, pn = p such that pi+1 is directlydensity-reachable from pi**
- **Density-connected**
  - **A point p is density-connected to a point q wrt. Eps, MinPts if there is a point o such that both, p and q are density-reachable from o wrt. Eps and MinPts.**

DBSCAN: Density Based Spatial Clustering of Applications with Noise :

- **Relies on a density-based notion of cluster: A cluster is defined as a maximalset of density-connected points**
- **Discovers clusters of arbitrary shape in spatial databases with noise**

DBSCAN: The Algorithm :

- **Arbitrary select a point p**
- **Retrieve all points density-reachable from p wrt Eps and MinPts.**
- **If p is a core point, a cluster is formed.**
- **If p is a border point, no points are density-reachable from p and DBSCANvisits the next point of the database.**
- **Continue the process until all of the points have been processed**

OPTICS: A Cluster-Ordering Method (1999) :

- **OPTICS: Ordering Points To Identify the Clustering Structure**
  - **Ankerst, Breunig, Kriegel, and Sander (SIGMOD'99)**
    - **Produces a special order of the database wrt its density-based clustering structure**
    - **This cluster-ordering contains info equiv to the density-based clusterings corresponding to a broad range of parameter settings**

- **Good for both automatic and interactive cluster analysis, includingfinding intrinsic clustering structure**
- **Can be represented graphically or using visualization techniques**

OPTICS: Some Extension from DBSCAN :

- **Index-based:**
  - **k = number of dimensions**
  - **N = 20**
  - **p = 75%**
  - **M = N(1-p) = 5**
- **Complexity: O(kN2)**
- **Core Distance**
- **Reachability Distance**

Max (core-distance (o), d (o, p))
r(p1, o) = 2.8cm. r(p2,o) = 4cm

DENCLUE: using density functions:

- **DENsity-based CLUstEring by Hinneburg & Keim (KDD'98)**
- **Major features**
  - **Solid mathematical foundation**
  - **Good for data sets with large amounts of noise**
  - **Allows a compact mathematical description of arbitrarily shapedclusters in high-dimensional data sets**
  - **Significant faster than existing algorithm (faster than DBSCAN by afactor of up to 45)**
  - **But needs a large number of parameters**

Denclue: Technical Essence:

- **Uses grid cells but only keeps information about grid cells that do actuallycontain data points and manages these cells in a tree-based access structure.**
- **Influence function: describes the impact of a data point within its neighborhood.**
- **Overall density of the data space can be calculated as the sum of the influencefunction of all data points.**

- **Clusters can be determined mathematically by identifying density attractors.**
- **Density attractors are local maximal of the overall density function.**

Grid-Based Methods:

Using multi-resolution grid data structure:

- **Several interesting methods**
  - **STING (a STatistical INformation Grid approach) by Wang, Yang andMuntz (1997)**
  - **WaveCluster by Sheikholeslami, Chatterjee, and Zhang (VLDB'98)**
    - **A multi-resolution clustering approach using wavelet method**
  - **CLIQUE: Agrawal, et al. (SIGMOD'98)**

STING: A Statistical Information Grid Approach:

- **Wang, Yang and Muntz (VLDB'97)**
- **The spatial area is divided into rectangular cells**
- **There are several levels of cells corresponding to different levels of resolution**

STING: A Statistical Information Grid Approach (2) :

- **Each cell at a high level is partitioned into a number of smaller cells inthe next lower level**
- **Statistical info of each cell is calculated and stored beforehand and isused to answer queries**
- **Parameters of higher level cells can be easily calculated from parameters of lower level cell**
  - **count, mean, s, min, max**
  - **type of distribution—normal, uniform, etc.**
- **Use a top-down approach to answer spatial data queries**
- **Start from a pre-selected layer—typically with a small number of cells**
- **For each cell in the current level compute the confidence interval**

STING: A Statistical Information Grid Approach (3) :

- **Remove the irrelevant cells from further consideration**
- **When finish examining the current layer, proceed to the next lowerlevel**
- **Repeat this process until the bottom layer is reached**
- **Advantages:**
  - **Query-independent, easy to parallelize, incremental update**
  - **O(K), where K is the number of grid cells at the lowest level**

- **Disadvantages:**
  - **All the cluster boundaries are either horizontal or vertical, andno diagonal boundary is detected**

WaveCluster (1998):

- **Sheikholeslami, Chatterjee, and Zhang (VLDB'98)**
- **A multi-resolution clustering approach which applies wavelet transform to thefeature space**
  - **A wavelet transform is a signal processing technique that decomposesa signal into different frequency sub-band.**
- **Both grid-based and density-based**
- **Input parameters:**
  - **# of grid cells for each dimension**
  - **the wavelet, and the # of applications of wavelet transform.**

- **How to apply wavelet transform to find clusters :**

  - **Summaries the data by imposing a multidimensional grid structureonto data space**
  - **These multidimensional spatial data objects are represented in a n-dimensional feature space**
  - **Apply wavelet transform on feature space to find the dense regions inthe feature space**
  - **Apply wavelet transform multiple times which result in clusters atdifferent scales from fine to coarse**

- **Why is wavelet transformation useful for clustering :**
  - **Unsupervised clustering**

It uses hat-shape filters to emphasize region where points cluster, but simultaneously to suppress weaker information in their boundary

  - **Effective removal of outliers**
  - **Multi-resolution**
  - **Cost efficiency**
- **Major features:**
  - **Complexity O(N)**
  - **Detect arbitrary shaped clusters at different scales**
  - **Not sensitive to noise, not sensitive to input order**

- **Only applicable to low dimensional data**
-

Model-Based Clustering Methods :

1. **Attempt to optimize the fit between the data and some mathematical model**
2. **Statistical and AI approachConceptual clustering**
3. **A form of clustering in machine learning**
4. **Produces a classification scheme for a set of unlabeled objects**
5. **Finds characteristic description for each concept (class)COBWEB (Fisher'87)**
6. **A popular a simple method of incremental conceptual learning**
7. **Creates a hierarchical clustering in the form of a classification tree**
8. **Each node refers to a concept and contains a probabilistic description of thatconcept**

Other Model-Based Clustering Methods:

1. Neural network approaches:

a. **Represent each cluster as an exemplar, acting as a "prototype" ofthe cluster**
b. **New objects are distributed to the cluster whose exemplar is themost similar according to some distance measure**

2. Competitive learning:

a. **Involves a hierarchical architecture of several units (neurons)**
b. **Neurons compete in a "winner-takes-all" fashion for the objectcurrently being presented**

CLIQUE (Clustering In QUEst):

- **Agrawal, Gehrke, Gunopulos, Raghavan (SIGMOD'98).**
- **Automatically identifying subspaces of a high dimensional data space thatallow better clustering than original space**
- **CLIQUE can be considered as both density-based and grid-based**
  - **It partitions each dimension into the same number of equal lengthinterval**
  - **It partitions an m-dimensional data space into non-overlappingrectangular units**
  - **A unit is dense if the fraction of total data points contained in the unitexceeds the input model parameter**
  - **A cluster is a maximal set of connected dense units within a subspace**

CLIQUE: The Major Steps :

- **Partition the data space and find the number of points that lie inside each cellof the partition.**
- **Identify the subspaces that contain clusters using the Apriori principle**
- **Identify clusters:**
  - **Determine dense units in all subspaces of interests**
  - **Determine connected dense units in all subspaces of interests.**
- **Generate minimal description for the clusters**
  - **Determine maximal regions that cover a cluster of connected denseunits for each cluster**
  - **Determination of minimal cover for each cluster**

Strength and Weakness of CLIQUE :

- **Strength**
  - **It automatically finds subspaces of the highest dimensionality suchthat high density clusters exist in those subspaces**
  - **It is insensitive to the order of records in input and does not presumesome canonical data distribution**
  - **It scales linearly with the size of input and has good scalability as thenumber of dimensions in the data increases**
- **Weakness**
  - **The accuracy of the clustering result may be degraded at the**

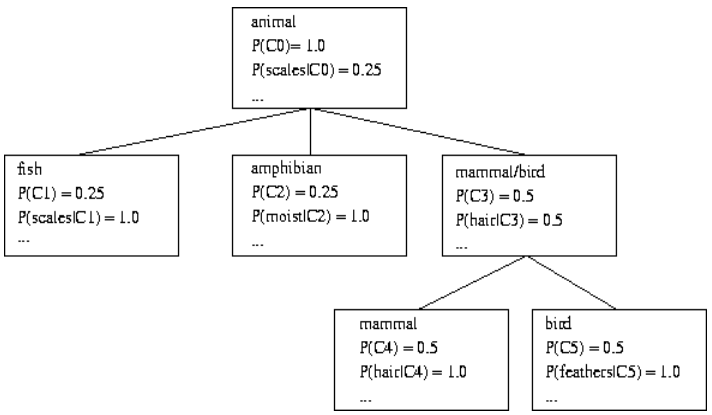
expenseof simplicity of the method

Model-Based Clustering Methods

- **Attempt to optimize the fit between the data and some mathematical model**
- **Statistical and AI approach**
  - **Conceptual clustering**
    - **A form of clustering in machine learning**
    - **Produces a classification scheme for a set of unlabeled objects**
    - **Finds characteristic description for each concept (class)**
  - **COBWEB (Fisher'87)**
    - **A popular a simple method of incremental conceptual learning**
    - **Creates a hierarchical clustering in the form of a classificationtree**
    - **Each node refers to a concept and contains a probabilistic description of that concept**

COBWEB Clustering Method :

A classification tree



More on Statistical-Based Clustering :

- **Limitations of COBWEB**
  - **The assumption that the attributes are independent of each**

other isoften too strong because correlation may exist

- **Not suitable for clustering large database data – skewed tree and expensive probability distributions**

- **CLASSIT**
  - **an extension of COBWEB for incremental clustering of continuousdata**
  - **suffers similar problems as COBWEB**
- **AutoClass (Cheeseman and Stutz, 1996)**
  - **Uses Bayesian statistical analysis to estimate the number of clusters**
  - **Popular in industry**
  -

Other Model-Based Clustering Methods :

- **Neural network approaches :**
  - **Represent each cluster as an exemplar, acting as a "prototype" of thecluster**
  - **New objects are distributed to the cluster whose exemplar is the mostsimilar according to some distance measure.**
- **Competitive learning :**
  - **Involves a hierarchical architecture of several units (neurons)**
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Clustering High-Dimensional Data in Data Mining :

Clustering is basically a type of unsupervised learning method. An unsupervised learning method is a method in which we draw references from datasets consisting of input data without labeled responses.

Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups.

Challenges of Clustering High-Dimensional Data:

Clustering of the High-Dimensional Data return the group of objects which are clusters. It is required to group similar types of objects together to perform the cluster analysis of high-dimensional data, But the High-Dimensional data space is huge and it has complex data types and attributes. A major challenge is that we need to find out the set of attributes that are present in each cluster. A cluster is defined and characterized based on the attributes present in the cluster. Clustering High-Dimensional Data we need to search for clusters and find out the space for the existing clusters.

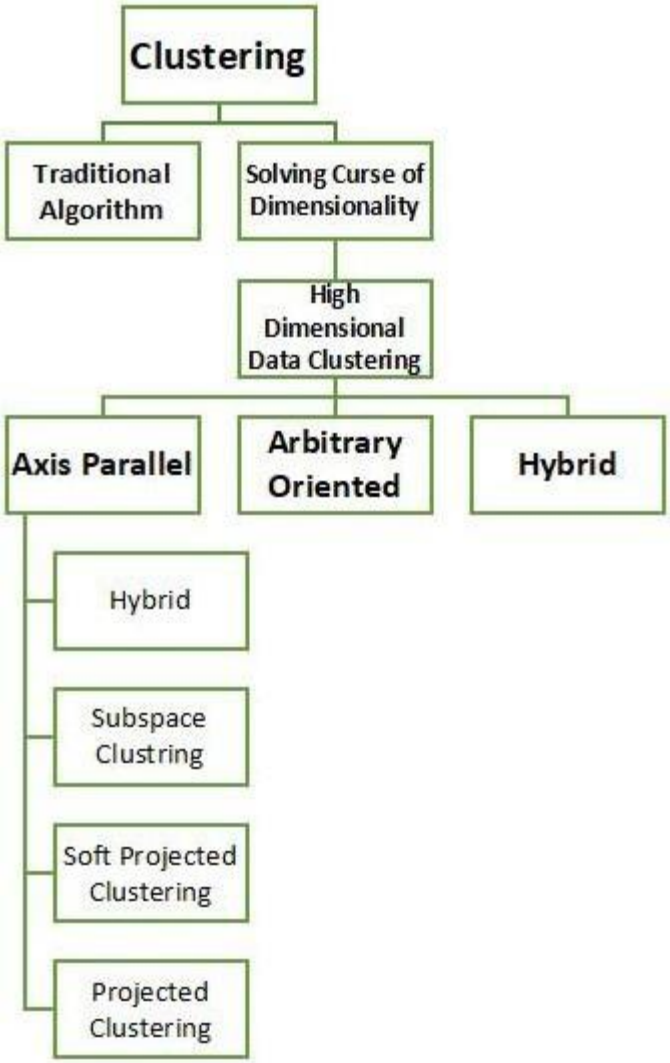
The High-Dimensional data is reduced to low-dimension data to make the clustering and search for clusters simple. some applications need the appropriate models of clusters, especially the high-dimensional data. clusters in the high-dimensional data are significantly small. the conventional distance measures can be ineffective. Instead, To find the hidden clusters in high-dimensional data we need to apply sophisticated techniques that can model correlations among the objects in subspaces.

Subspace Clustering Methods:

There are 3 Subspace Clustering Methods:

- **Subspace search methods**
- **Correlation-based clustering methods**
- **Biclustering methods**

Subspace clustering approaches to search for clusters existing in subspaces of the given high-dimensional data space, where a subspace is defined using a subset of attributes in the full space.



1. Subspace Search Methods: A subspace search method searches the subspaces for clusters. Here, the cluster is a group of similar types of objects in a subspace. The similarity between the clusters is measured by using distance or density features. CLIQUE algorithm is a subspace clustering method. subspace search methods search a series of subspaces. There are two approaches in Subspace Search Methods: Bottom-up approach starts to search from the low-dimensional subspaces. If the hidden clusters are not found in

low-dimensional subspaces then it searches in higher dimensional subspaces. The top-down approach starts to search from the high-dimensional subspaces and then search in subsets of low-dimensional subspaces. Top-down approaches are effective if the subspace of a cluster can be defined by the local neighborhood sub-space clusters.

2. Correlation-Based Clustering: correlation-based approaches discover the hidden clusters by developing advanced correlation models. Correlation-Based models are preferred if is not possible to cluster the objects by using the Subspace Search Methods. Correlation-Based clustering includes the advanced mining techniques for correlation cluster analysis. Biclustering Methods are the Correlation-Based clustering methods in which both the objects and attributes are clustered.

3. Biclustering Methods:

Biclustering means clustering the data based on the two factors. we can cluster both objects and attributes at a time in some applications. The resultant clusters are biclusters. To perform the biclustering there are four requirements:

- **Only a small set of objects participate in a cluster.**
- **A cluster only involves a small number of attributes.**
- **The data objects can take part in multiple clusters, or the objects may also include in any cluster.**
- **An attribute may be involved in multiple clusters.**

Objects and attributes are not treated in the same way. Objects are clustered according to their attribute values. We treat Objects and attributes as different in biclustering analysis.

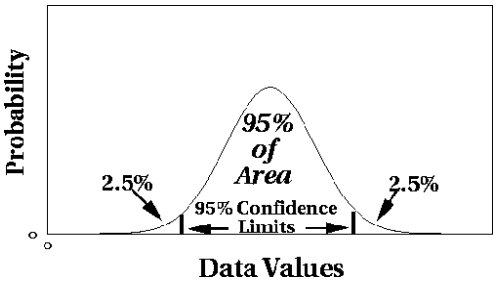
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Outlier Analysis :

What Is Outlier Discovery?

- **What are outliers?**
  - **The set of objects are considerably dissimilar from the remainder ofthe data**
  - **Example: Sports: Michael Jordon, Wayne Gretzky, ...**
- **Problem**
  - **Find top n outlier points**
- **Applications:**
  - **Credit card fraud detection**
  - **Telecom fraud detection**
  - **Customer segmentation**
  - **Medical analysis**

Outlier Discovery: Statistical Approaches :



- **Assume a model underlying distribution that generates data set (e.g. normal distribution)**
- **Use discordance tests depending on**
  - **data distribution**
  - **distribution parameter (e.g., mean, variance)**
  - **number of expected outliers**
- **Drawbacks**
  - **most tests are for single attribute**
  - **In many cases, data distribution may not be known**

Outlier Discovery: Distance-Based Approach :

- **Introduced to counter the main limitations imposed by statistical methods**
  - **We need multi-dimensional analysis without knowing datadistribution.**
- **Distance-based outlier: A DB(p, D)-outlier is an object O in a dataset T**

such that at least a fraction p of the objects in T lies at a distance greater than D from O

- **Algorithms for mining distance-based outliers**
  - **Index-based algorithm**
  - **Nested-loop algorithm**
  - **Cell-based algorithm**
  -

Outlier Discovery: Deviation-Based Approach :

- **Identifies outliers by examining the main characteristics of objects in a group Objects that "deviate" from this description are considered outliers**
- **sequential exception technique**
  - **simulates the way in which humans can distinguish unusual objectsfrom among a series of supposedly like objects**
- **OLAP data cube technique**
  - **uses data cubes to identify regions of anomalies in largemultidimensional data**

Applications of data mining :

- **Data mining is a young discipline with wide and diverse applications**
  - **There is still a nontrivial gap between general principles of data mining and domain-specific, effective data mining tools for particular applications**
- **Some application domains (covered in this chapter)**
  - **Biomedical and DNA data analysis**
  - **Financial data analysis**
  - **Retail industry**
  - **Telecommunication industry**

Biomedical Data Mining and DNA Analysis :

- **DNA sequences: 4 basic building blocks (nucleotides): adenine (A), cytosine(C), guanine (G), and thymine (T).**
- **Gene: a sequence of hundreds of individual nucleotides arranged in a particular order**
- **Humans have around 100,000 genes**
- **Tremendous number of ways that the nucleotides can be ordered and sequenced to form distinct genes**
- **Semantic integration of heterogeneous, distributed genome databases**

- **Current: highly distributed, uncontrolled generation and use of a widevariety of DNA data**
- **Data cleaning and data integration methods developed in data miningwill help**
-

DNA Analysis: Examples :

- **Similarity search and comparison among DNA sequences**
  - **Compare the frequently occurring patterns of each class (e.g., diseasedand healthy)**
  - **Identify gene sequence patterns that play roles in various diseases**
- **Association analysis: identification of co-occurring gene sequences**
  - **Most diseases are not triggered by a single gene but by a combinationof genes acting together.**
  - **Association analysis may help determine the kinds of genes that arelikely to co-occur together in target samples**
- **Path analysis: linking genes to different disease development stages**
  - **Different genes may become active at different stages of the disease**
  - **Develop pharmaceutical interventions that target the different stagesseparately**
- **Visualization tools and genetic data analysis**

Data Mining for Financial Data Analysis :

- **Financial data collected in banks and financial institutions are often relativelycomplete, reliable, and of high quality**
- **Design and construction of data warehouses for multidimensional data analysis and data mining**
  - **View the debt and revenue changes by month, by region, by sector,and by other factors**
  - **Access statistical information such as max, min, total, average, trend,etc.**
- **Loan payment prediction/consumer credit policy analysis**
  - **feature selection and attribute relevance ranking**
  - **Loan payment performance**
  - **Consumer credit rating**

Financial Data Mining :

- **Classification and clustering of customers for targeted marketing**

- **multidimensional segmentation by nearest-neighbor, classification, decision trees, etc. to identify customer groups or associate a new customer to an appropriate customer group**

- **Detection of money laundering and other financial crimes**
  - **integration of from multiple DBs (e.g., bank transactions, federal/statecrime history DBs)**
  - **Tools: data visualization, linkage analysis, classification, clustering tools, outlier analysis, and sequential pattern analysis tools (find unusual access sequences)**

Data Mining for Retail Industry :

- **Retail industry: huge amounts of data on sales, customer shopping history,etc.**
- **Applications of retail data mining**
  - **Identify customer buying behaviors**
  - **Discover customer shopping patterns and trends**
  - **Improve the quality of customer service**
  - **Achieve better customer retention and satisfaction**
  - **Enhance goods consumption ratios**
  - **Design more effective goods transportation and distribution policies**

Data Mining in Retail Industry: Examples :

- **Design and construction of data warehouses based on the benefits of datamining**
  - **Multidimensional analysis of sales, customers, products, time, and region**
- **Analysis of the effectiveness of sales campaigns**
- **Customer retention: Analysis of customer loyalty**
  - **Use customer loyalty card information to register sequences of purchases of particular customers**
  - **Use sequential pattern mining to investigate changes in customer consumption or loyalty**
  - **Suggest adjustments on the pricing and variety of goods**
- **Purchase recommendation and cross-reference of items**

Data Mining for Telecomm. Industry :

- **A rapidly expanding and highly competitive industry and a great demand fordata mining**

  - **Understand the business involved**
  - **Identify telecommunication patterns**
  - **Catch fraudulent activities**
  - **Make better use of resources**
  - **Improve the quality of service**
- **Multidimensional analysis of telecommunication data**
  - **Intrinsically multidimensional: calling-time, duration, location of caller, location of callee, type of call, etc.**
- **Fraudulent pattern analysis and the identification of unusual patterns**
  - **Identify potentially fraudulent users and their atypical usage patterns**
  - **Detect attempts to gain fraudulent entry to customer accounts**
  - **Discover unusual patterns which may need special attention**
- **Multidimensional association and sequential pattern analysis**
  - **Find usage patterns for a set of communication services by customergroup, by month, etc.**
  - **Promote the sales of specific services**
  - **Improve the availability of particular services in a region**
- **Use of visualization tools in telecommunication data analysis.**
